AUTOMATED FISH SPECIES CLASSIFICATION USING SONAR AND MACHINE LEARNING

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BACKGROUND



- Ontario Government needs to perform ecological monitoring in Ontario lakes
  - To know what's in the lake
  - Approximate fish population
  - Provide guidance for the inland fisheries

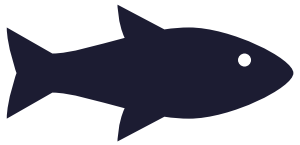
Picture sources from Toronto and Region Conservation Authority

EXISTING METHODS



Trawling / Netting

- Towing a large, cone-shaped net (a trawl) through the water to catch fish
- Invasive and harm fish population
- Time consuming and expensive
- Limit large-scale or frequent monitoring



Hydroacoustic

- Uses sonar technology to track and assess fish populations and their behavior by analyzing sound waves reflected off fish
- Non-invasive
- Rely on manual interpretation

RESEARCH QUESTIONS

- Developed automated workflow to implement hydroacoustic with Machine Learning techniques
  - Build Machine Learning models to automatically classify fish species using labeled data
  - Focus on fish-level classification at the moment

DATA - RAW

- 14574 observations, 484 variables
- 21 Lake Trout, 11 Lake Whitefish, 12 Smallmouth Bass

Fish Characteristics | Target Strength

FishNum	species	totalLength	...	pingNumber	F45	...	...	F260
LT001	lakeTrout			1				
LT001	lakeTrout			2				
LT001	lakeTrout			3				
...	...			...				
SMB013	smallmouthBass							

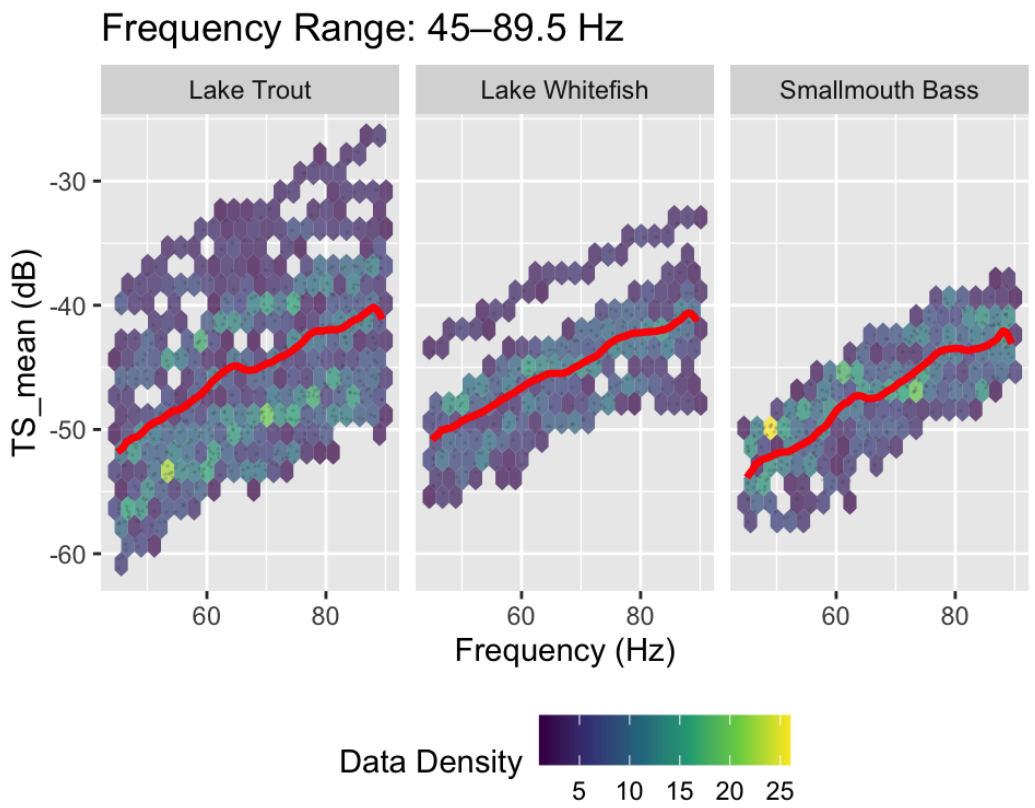
DATA - PROCESSED

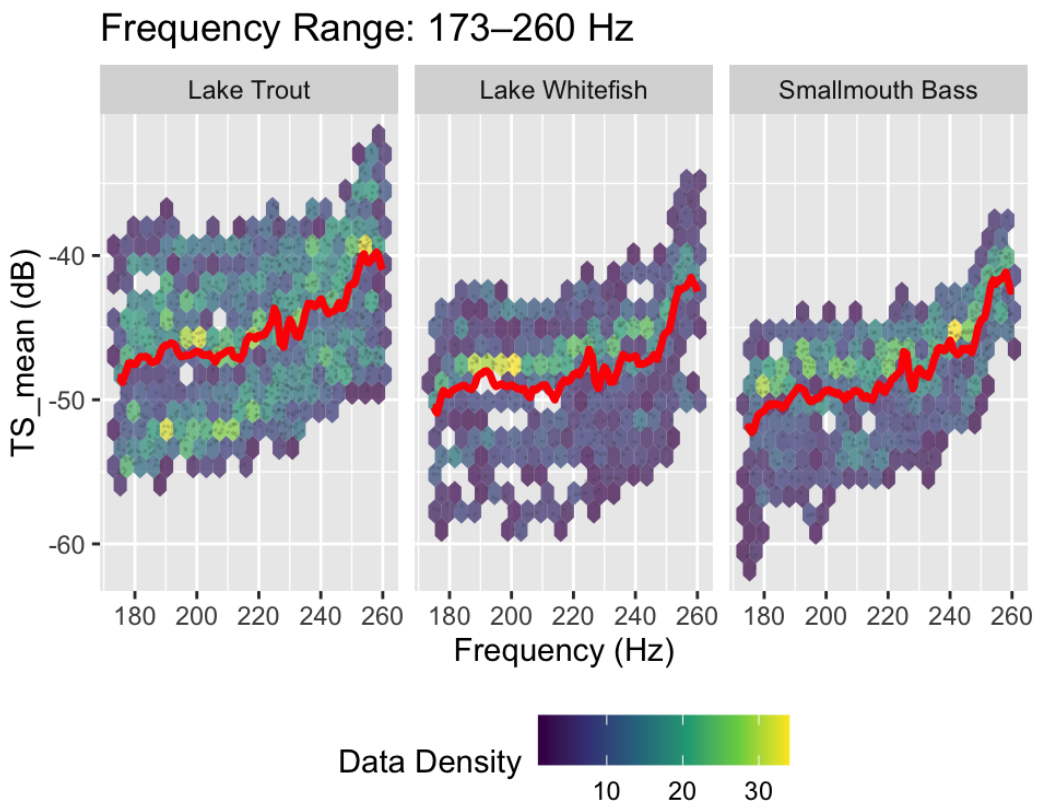
- 14392 observations, 264 variables

- 21 Lake Trout, 10 Lake Whitefish, 12 Smallmouth Bass

FishNum	species	F45	...	F89.5	F173	...	...	F260
LT001	lakeTrout							
LT001	lakeTrout							
LT001	lakeTrout							
...	...							
SMB013	smallmouthBass							

DATA EXPLORATION





PROPOSED METHODS

Multinomial Regression

- Simple baseline model to test linear relationship in the data
- Provide interpretable results and computationally lightweight

Random Forest

- Handle hierarchical structure of high-dimensional fish signatures
- Robust to noise and overfitting, reveal feature importance

Deep Neural Network

- Learn complex, non-linear hierarchical interactions
- Automatic layered abstraction

LSTM (Long Short-Term Memory)

- Designed for sequential data
- Incorporate fish habitat context for temporal patterns

PROPOSED METHODS – INPUT DATA

Multinomial Regression

- Mean TS for each fish
- Standardized and apply PCA (first two PCs)
- Dim: 43 x 267

FishNum	species	F45	...	F89.5	F173	...	F260
LT001	lakeTrout	(mean)	(mean)	(mean)	(mean)	(mean)	(mean)
LT002	lakeTrout	...	...	...	...	...	...
...	...						
SMB013	smallmouthBass						

Random Forest

- Raw data
- Dim: 14392 x 267

FishNum	species	F45	...	F89.5	F173	...	F260
LT001	lakeTrout						
LT001	lakeTrout						
...	...						
SMB013	smallmouthBass						

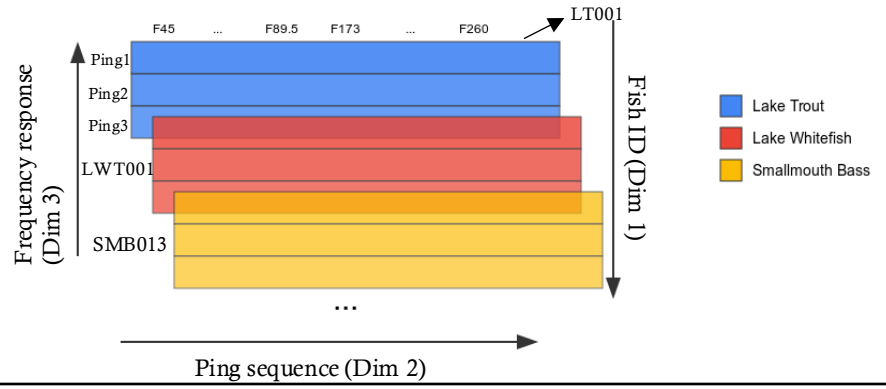
Deep Neural Network

- Raw data
- Dim: 14392 x 267

FishNum	species	F45	...	F89.5	F173	...	F260
LT001	lakeTrout						
LT001	lakeTrout						
...	...						
SMB013	smallmouthBass						

LSTM

- Standardized 3D array
- Dim: 43 x 1619 (N) x 267



RESULTS

Multinomial Regression

- Accuracy: 37.5%
- Apply PCA and uses first two PCs (Cum Var: 95.9%)

Actual Predicted	LT	LWT	SMB
LT	3	1	2
LWT	0	0	0
SMB	1	1	0

Random Forest

- Accuracy: 62.5%
- Mtry: 20, n_Trees: 400, min_n: 5

Actual Predicted	LT	LWT	SMB
LT	4	1	2
LWT	0	1	0
SMB	0	0	0

Deep Neural Network

- Accuracy: 62.5%
- 3 hidden layer, epochs: 128

Actual Predicted	LT	LWT	SMB
LT	4	1	2
LWT	0	1	0
SMB	0	0	0

LSTM

- Accuracy: 62.5%
- 2 LSTM layers, 1 hidden layer, epochs: 21

Actual Predicted	LT	LWT	SMB
LT	2	0	1
LWT	1	2	0
SMB	1	0	1

FUTURE STEPS

- Enhanced LSTM with attention mechanism
  - Focus on most informative signal
  - Less focus on the less important signal

- Incorporate Fish characteristics
  - Signal variation within species might relate to weight, total length, gender, etc.

More details can be view at GitHub: Fish Classification

THANK YOU